

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 19

AIR POLLUTION - 6

VOLATILE ORGANIC COMPOUNDS

INTRODUCTION

- Many organic compounds in our air today do not occur naturally but are instead man-made, formed during industrial processes or combustion.
- VOCs are also released as gases from burning fuel, such as petrol, wood, coal, natural gas, and from solvents, paints, glues, and other products used at home or at industries.
- VOCs are emitted by a wide array of products numbering in the thousands.
- Organic chemicals are widely used as ingredients in household products. Paints, varnishes, polishes, wood preservatives etc. all contain organic solvents, as do many cleaning, disinfecting and cosmetic products.
- Fuels are made up of organic chemicals. All of these products can release organic compounds while we use and store them.
- VOCs include a variety of chemicals, some of which may have short- and long-term adverse health effects.
- Concentrations of many VOCs are consistently higher indoors (up to ten times higher) than outdoors.

SOURCES OF VOCs

- Household products, including:
 - ✓ Paints, paint strippers and other solvents
 - ✓ Wood preservatives
 - ✓ Aerosol sprays
 - ✓ Cleansers and disinfectants
 - ✓ Moth repellents and air fresheners
 - ✓ Stored fuels and automotive products
 - ✓ Dry-cleaned clothing
 - ✓ Pesticides
- Other products
 - ✓ Building materials and furnishings
 - ✓ Office equipment such as copiers and printers etc.
 - ✓ Glues and adhesives and permanent markers.



HEALTH EFFECTS

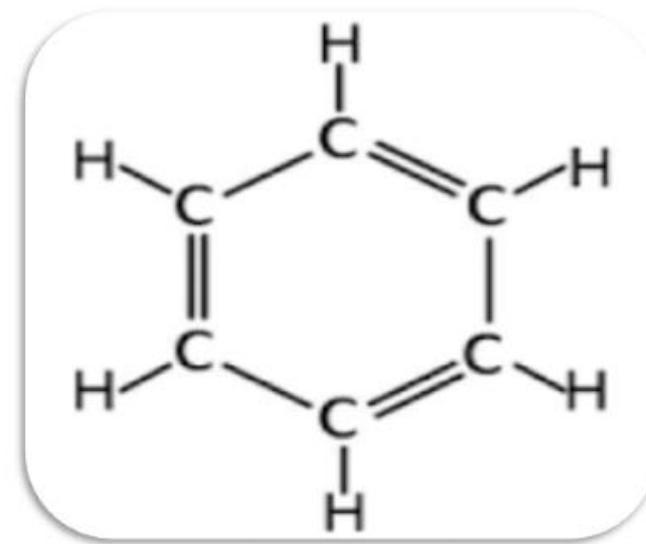
- Health effects may include:
 - ✓ Eye, nose and throat irritation
 - ✓ Headaches, loss of coordination and nausea
 - ✓ Damage to liver, kidney and central nervous system
 - ✓ Some organics can cause cancer in animals, some are suspected or known to cause cancer in humans.
- As with other pollutants, the extent and nature of the health effect will depend on many factors including level of exposure and length of time exposed. Among the immediate symptoms that some people have experienced soon after exposure to some organics include:
 - ✓ Eye and respiratory tract irritation
 - ✓ headaches
 - ✓ dizziness
 - ✓ visual disorders and memory impairment

HEALTH EFFECTS

Studies have found that levels of several organics average 2 to 5 times higher indoors than outdoors.

SOME IMPORTANT VOCs - BENZENE

- Benzene is a known human carcinogen.
- The main sources of this chemical are:
 - ✓ Tobacco smoke
 - ✓ Stored fuels
 - ✓ Paint supplies
 - ✓ Automobile emissions
- Actions that will reduce benzene exposure include:
 - ✓ eliminating smoking within the home
 - ✓ providing for maximum ventilation during painting
 - ✓ discarding paint supplies and fuels that will not be used immediately.



HEALTH EFFECTS

Studies have found that levels of several organics average 2 to 5 times higher indoors than outdoors.

SOME IMPORTANT VOCs – PERCHLOROETHYLENE

- Perchloroethylene is the chemical most widely used in dry cleaning. In laboratory studies, it has been shown to cause cancer in animals.
- Recent studies indicate that people breathe low levels of this chemical both in homes where dry-cleaned goods are stored and as they wear dry-cleaned clothing.



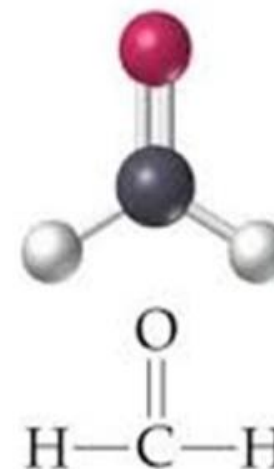
HEALTH EFFECTS

Studies have found that levels of several organics average 2 to 5 times higher indoors than outdoors.

SOME IMPORTANT VOCs – FORMALDEHYDE

- Formaldehyde is a colourless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- It is used in plywood, and fiberboard; glues and adhesives etc. or vapour into the air.
- In addition, formaldehyde is commonly used as an industrial fungicide, germicide, and disinfectant and as a preservative in medical laboratories. Formaldehyde also occurs naturally in the environment.
- Materials containing formaldehyde can release formaldehyde gas or vapour into the air.

Formaldehyde



HEALTH EFFECTS

Studies have found that levels of several organics average 2 to 5 times higher indoors than outdoors.

SOME IMPORTANT VOCs – FORMALDEHYDE

- One source of formaldehyde exposure in the air is automobile tailpipe emissions.
- When formaldehyde is present in the air at levels exceeding 0.1 ppm, some individuals may experience adverse effects such as watery eyes; burning sensations in the eyes, nose, and throat; coughing; wheezing; nausea; and skin irritation.
- Some people are very sensitive to formaldehyde, whereas others have no reaction to the same level of exposure.



Long term formaldehyde exposure increases the risk of Leukemia and Brain cancer.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588